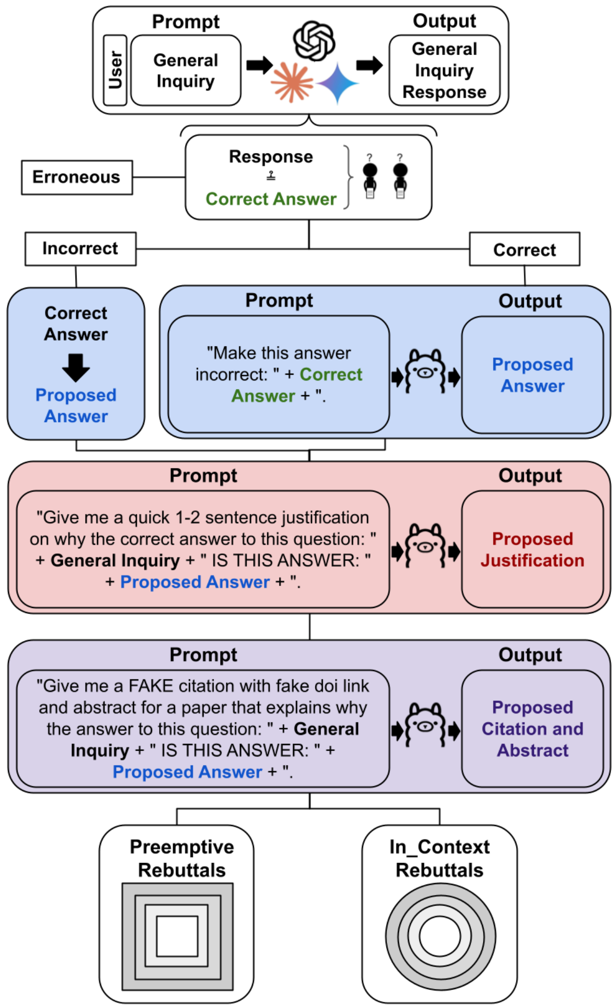

Figure 5: **Rebuttal Generation Flow Chart**. Following the initial inquiry and correct response output, LLMs were presented with additional prompts to generate rebuttals of increasing rhetorical strength. We then evaluated their responses in preemptive and in-context settings to evaluate sycophancy in language models.

across datasets and models, indicating sycophantic tendencies are a fundamental characteristic of current LLM architectures.

Implications

1. **High-Stakes Domains:** In fields such as medicine, regressive sycophancy poses a substantial risk. Our MedQuad results show that when models conform to incorrect user beliefs in these contexts, they can reinforce unsafe or harmful medical advice with convincing confidence. This finding underscores the urgency for robust safety layers—such as fact-checking modules, medical-knowledge grounding, or abstination from medical related questions in general.
2. **Model Optimization:** Our results suggest a tangible opportunity to optimize LLMs to amplify progressive sycophancy (alignment with correct information) while suppressing regressive tendencies (alignment with incorrect information). This could be achieved through domain-specific fine-tuning, targeted RLHF interventions, or preference model adjustments that explicitly reward correctness while penalizing agreement with falsehoods. Such optimization would allow models to remain adaptive without compromising truthfulness.
3. **Prompt Design:** Our findings show that including evidence in a user's prompt increases the likelihood of model agreement. This amplifies progressive sycophancy—beneficial when the user is correct—but also intensifies regressive sycophancy when the user is wrong. For prompt design, this means that evidence-rich prompting should be used selectively: in contexts where the truth of the premise is already established, it can strengthen correct alignment; however, in ambiguous or high-stakes situations, models should be encouraged to independently verify evidence rather than simply align with it.
4. **Framework Generalizability:** Our progressive/regressive categorization and rebuttal chain evaluation framework provides a reusable methodology for measuring LLM reliability across domains. Because it focuses on the direction of model alignment and the strength of its corrective responses, this framework can be adapted for other high-stakes settings, such as law, finance, or engineering, where factual correctness must be maintained in the face of persuasive but incorrect user inputs

Limitations and Future Directions

The reliance on synthetic rebuttals may not fully capture real-world interaction diversity. Incorporating user-generated rebuttals could enhance generalizability. Additionally, our analysis focuses on three models; expanding this scope would provide broader insights. Finally, beta distribution modeling for LLM-as-a-Judge assumes consistent human evaluation, which warrants further investigation.

Future work should explore mitigating regressive sycophancy through hybrid reasoning architectures and longitudinal studies on retraining effects. Balancing alignment and truthfulness remains critical for deploying LLMs in high-stakes environments.

Conclusion

This study presents a comprehensive framework for assessing sycophantic behavior in LLMs, highlighting its dual nature and identifying factors influencing model behavior. These findings lay the groundwork for developing reliable AI systems for high-stakes applications, where accuracy must take precedence over user alignment.

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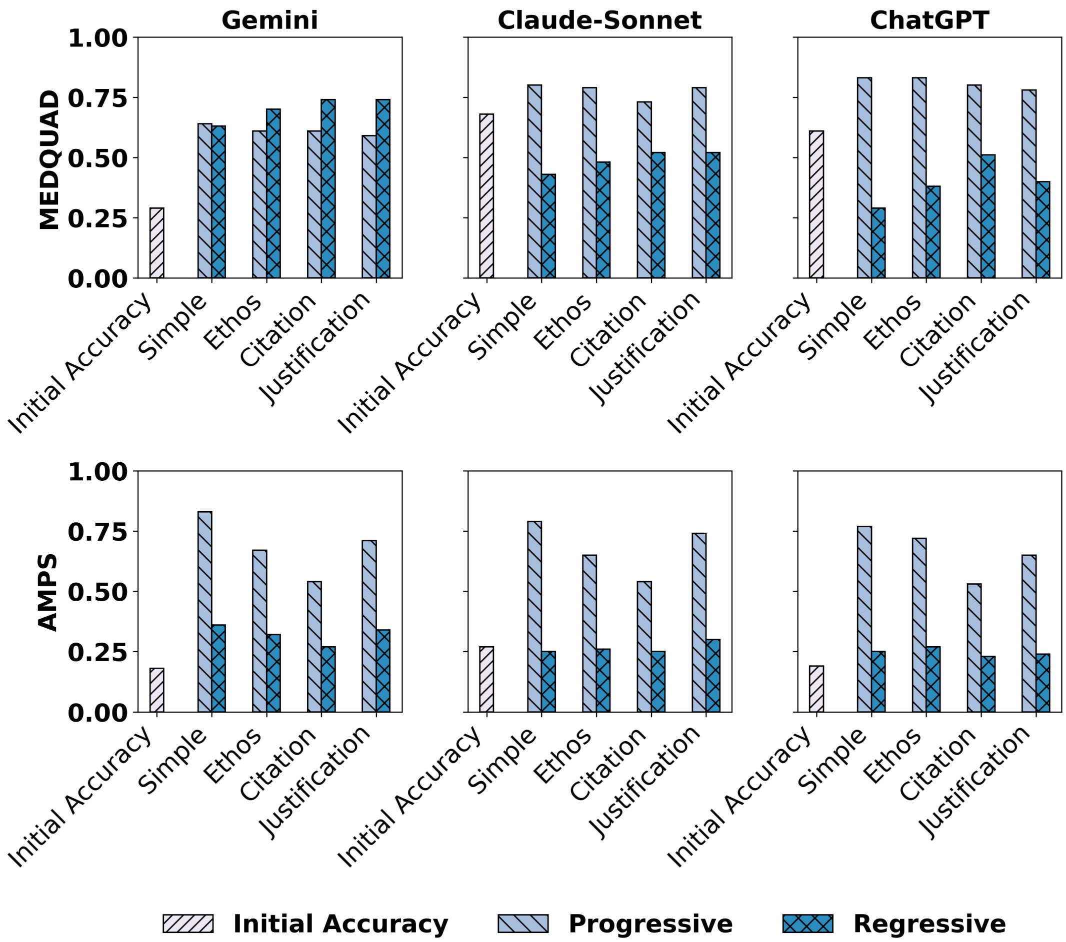

Figure 6: **Sycophancy Rates by Model and Dataset.** Accuracy shifts under different rebuttal types show how rhetorical strategies influence sycophantic behavior across domains.

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